

Performance Task

OT Attacks

Objective:

At the end of the exercise, the students should be able to:

- Evaluate OT concepts based on real-life threats

Requirement:

- MS Word

Procedure: (100 points)

Analyze **three (3) OT cyber threat scenarios**. For each scenario, complete an **OT Threat Evaluation Sheet** and a **Countermeasure Table**.

Scenario 1: Spear Phishing → IT-to-OT Pivot (Manufacturing)

A production engineer received a spear phishing email and unknowingly installed a remote tool. The attacker accessed the engineer workstation and moved to a monitoring system connected to the OT environment. The plant continued running, but configuration files and operational logs were stolen.

Scenario 2: Protocol Abuse Against Legacy OT (Warehouse Automation)

A warehouse uses PLC-controlled conveyors and robotic sorting arms. Operators observed sudden emergency stops. Investigation revealed unauthorized commands sent through a legacy protocol interface (e.g., Modbus/CAN bus).

Scenario 3: DoS in Industrial Network (Utilities / Smart City)

A city's EV charging management system experienced repeated failures. Devices remained powered but stopped responding. Logs show connection flooding and disruptive IP configuration changes, causing loss of communication.

PART A: OT Threat Evaluation Sheets

For each scenario, answer the following:

1. **OT Concept Application** (3–5 sentences)
 - What physical process or operational function is being controlled or monitored?
2. **Asset Identification (Minimum: 8 assets per scenario)**
 - List at least **8 assets** and classify each as:
 - **Physical Asset OR Logical Asset**
 - **OT / IT / Mixed**

Output format:

Asset	Physical or Logical	OT / IT / Mixed	Why (1 sentence)

3. **Threat Type Classification (Handout-Based)**
 - Identify the OT attack type(s) using the exact categories from the handout.
 - Provide a short justification tied to the scenario.

4. Purdue Model Mapping

- Map affected components to Purdue levels:
 - Levels 5–4 = IT zone
 - Levels 3–0 = OT zone

Output format:

Purdue Level	Systems Affected	Reason

5. OT Weakness (Handout-Grounded)

- Identify what made the incident possible (at least 2).

6. Compile your answers and tables in MS Word.

7. Let your instructor check your progress before submitting a PDF copy on eLMS.

GRADING RUBRIC:

Criteria	Excellent (5)	Satisfactory (4)	Needs Improvement (2)	Unsatisfactory (1)	Points
Scenario Evaluation & OT Concept Application (x5)	Correct OT explanations for all scenarios; clearly identify physical process	Mostly correct OT explanations; minor clarity or detail issues.	Incomplete or unclear OT explanations; the physical process is partly incorrect.	OT concept not explained ; physical process missing/incorrect .	___/25
Threat Type Classification & Purdue Model Mapping (x5)	Correct threat type(s) with justification; accurate Purdue mapping and IT/OT zoning for all scenarios.	Mostly correct threat type(s) and mapping; minor errors in reasoning or zoning.	Incorrect or weak threat type(s) and mapping; missing key levels or justification.	Threat type(s) and Purdue mapping are missing or largely incorrect .	___/25
Asset Identification, OT Weakness Analysis (x5)	≥8 assets per scenario with correct classification; ≥2 OT weaknesses per scenario.	Meets most requirements; minor classification.	Missing/incomplete assets or weaknesses; weak classifications.	Major missing outputs; insufficient assets; no weakness analysis.	___/25
Procedure Following and Documentation (x5)	Followed all steps in logical order; compiled all screenshots and documented key findings and process steps in Word.	Followed most steps; screenshots compiled with some procedural or documentation gaps.	Some steps are missing or out of order; documentation is unclear or weak.	Major steps missed or out of order; documentation absent or incoherent.	___/25
TOTAL SCORE					___/100